

Harshit Katyal

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EDUCATION

- Thapar Institute of Engineering and Technology, Patiala** Aug 2023 – Aug 2027
B.E. in Computer Engineering (CGPA: 9.55/10)
- Merit-based scholarship (Top 10%)
- The SD Vidya School, Ambala Cantt** July 2022 – May 2023
Class 12 (CBSE) 95%
- Secured **2nd rank** in school; highest marks in **Physics (95)**
- The SD Vidya School, Ambala Cantt** July 2020 – May 2021
Class 10 (CBSE) 96%
- Consistently among the **top 5%** performers in academics

TECHNICAL SKILLS

Languages: C, C++, Python, R, MATLAB, HTML, CSS, JavaScript
Data Science: Pandas, NumPy, Scikit-learn, Matplotlib
Databases: Oracle, PL/SQL
Tools: Git, GitHub, VS Code, Google Colab
Core CS & Coursework: DSA, OOPS, OS, DBMS, CN, ML, DL, Predictive Statistics

PROJECTS

- Car Crash Simulation & Impact Prediction** | *Python, SimPy, Pandas, Scikit-Learn, Matplotlib* | [Source Code](#)
- Developed a SimPy-based vehicle braking and collision simulation framework, generating **1000** synthetic observations under varying road and driver conditions.
 - Built and benchmarked **7** regression models including Random Forest, Decision Tree, KNN, and SVR for impact-force prediction.
 - Evaluated models using R^2 , RMSE, and MAE metrics; Random Forest achieved the highest **R^2 score of 0.9686**.
- Sports Inventory System** | *PL/SQL, Oracle* | [Source Code](#)
- DBMS for equipment tracking with automated **10/day** fine calculation and stock alerts.
 - Developed procedures, triggers and cursors for student performance-based reporting.
 - Tracked **10+ entities** with **25+ procedures** across inventory, return, fine, and performance modules.
- TOPSIS Python Package & Ranking Platform** | *Python, Pandas, NumPy, PyPI, Next.js* | [Source Code](#)
- Developed and published a Python package on **PyPI** implementing the TOPSIS multi-criteria decision-making algorithm for ranking alternatives.
 - Built a web-based platform supporting **CSV/Excel** uploads, configurable weights and impacts, and automated TOPSIS-based ranking analysis.

POSITIONS OF RESPONSIBILITY

- **AIChE (Core Member, Design Team):** Created **15+** promotional assets for society events, increasing engagement by **25%**.
- **OWASP (Executive Member):** Led outreach and publicity for HackOWASP and CTF events, driving **350+** registrations and a **40%–70%** increase in participation.

ACHIEVEMENTS & CERTIFICATIONS

Achievements: **2nd** Position in Haryana (Fit India Quiz); Participated in SatHack (NVIDIA) and Hacklipse (ACM).

Certifications: [Pandas & NumPy](#), [SQL Analytics](#), [Advanced SQL](#), [Agentic AI](#).